

CSC110 — Constructing a Clinical Trial Cohort

Context

Clinical trials need to compare treatment effects in a well-defined population — if the group of subjects varies too much in disease severity, treatment history, or data completeness, it becomes hard to tell whether a difference in outcomes came from the treatment or from who happened to be in the study. Every trial therefore specifies a precise list of **inclusion criteria** (conditions a subject must meet to enroll) and **exclusion criteria** (conditions that disqualify them), and the *trial cohort* is exactly the set of real-world patients who satisfy all of the inclusion criteria and none of the exclusion criteria.

That’s exactly what you’ll build today: starting from 392 real (de-identified) follicular lymphoma patient records, you’ll construct the cohort for a hypothetical trial by applying a list of inclusion and exclusion criteria — using nothing but the if-structures, loops, and list features you already know.

The Data

You’ll work with `sample_patient_data.py`. Import it (or copy its contents into your script) to get access to parallel lists — `ids`, `regimen`, `ipi_score`, `disease_stage`, `n_lots`, `initial_diagnosis_date`, and others — where index i in every list refers to the same patient. See the data dictionary at the top of that file for a full column-by-column description.

The Trial

Scenario: you’re constructing the cohort for a hypothetical trial studying **higher-risk follicular lymphoma patients treated with bendamustine-based first-line therapy**.

Inclusion criteria — a subject qualifies only if they satisfy **ALL** of:

#	Criterion
I1	Initial diagnosis occurred in 2018 or later
I2	IPI score of 3 or higher at diagnosis
I3	First-line regimen included bendamustine

Exclusion criteria — a subject is dropped if they satisfy **ANY** of:

#	Criterion
E1	Disease stage was not recorded
E2	Received more than one line of therapy

Your Task

1. **Write one filtering expression per criterion first**, and print how many of the 392 subjects satisfy each one *individually*. This lets you sanity check each piece before combining them — if a criterion doesn't narrow the population at all, or drops everyone, something is probably wrong.
2. **Combine all five criteria** into a single final cohort — a list of the `ids` values for every subject who meets all three inclusion criteria and none of the exclusion criteria. You may write this either as a sequence of `if` checks inside a loop, or as a single list comprehension — both are fair game.
3. **Report your final cohort size.** (You should be able to check your answer with a classmate or the instructor.)

Hints

- I1 needs no date parsing — every date in this dataset is formatted MM/DD/YYYY, so the year is always the **last 4 characters** of the string. Slicing handles this directly: `date_string[-4:]`.
- I2 requires no type conversion — `ipi_score` is already a list of `int`.
- I3 is a list-membership check: is "bendamustine" one of the strings inside `regimen[i]`?
- E1 is a blank-string check on `disease_stage[i]`.
- E2 needs `int(...)` conversion, since `n_lots` is stored as strings.

Stretch

- Provide some summary statistics on your resulting cohort (e.g. mean age, distribution of stage 3 vs stage 4, male vs female counts)
- How many subjects would qualify if you *removed* the E2 criterion (allowed any number of treatment lines)? Was E2 doing much work?
- Design one additional, sensible exclusion criterion of your own, using a column not touched above (e.g. `sex`, `cr`, `os_censor_flag`), and report how much further it narrows the cohort.